

Ensemble Embedding Collapse under Standing-Preserving Admissibility

Transport, Localization, Operator Exhaustion, and the Uniqueness of the Admissible Interior

Amos Jay Maley

May 2026

Abstract

This paper proves an admissibility-theoretic ensemble-collapse theorem for standing-bearing comparison structures. The target is not metaphysical monism, logical uniqueness of possible worlds, or denial of formal plurality. The target is narrower: whether a family of purportedly independently standing-bearing admissibility domains can inherit a single admissible comparison regime merely by being collected as an ensemble. Building on fixed-domain identity closure, admissible-interior uniqueness, operator exhaustion, cross-domain closure, licensed transport, compatibility licensing, continuation-locus induction, multiplicity minimality, and bounded standing localization, we show that any purported ensemble of standing-bearing domains must resolve into one of a finite set of outcomes. Either its members collapse extensionally into the same admissible interior; their differences are presentation-level bookkeeping; the construction requires illicit scope transport; it imports standing without independent target gating; it relies on inadmissibility-relevant parameterization; it uses a selector, carrier, generator, or repair operator; it fails cross-domain overlap/translation discipline; it fails transport closure or compatibility; it lacks continuation-locus preservation; it relies on unbounded or locally indistinguishable multiplicity; or it fails bounded persistence localization. No inherited governance-relevant admissible ensemble multiplicity is conferred by multiplicity alone. Any surviving multiplicity must be retyped as an explicitly gated target domain, where it is then governed by the same admissibility, transport, localization, and closure constraints.

The theorem is stated in a scope-locked form. It preserves ordinary representational plurality, model plurality, coordinate plurality, and formal description plurality. What collapses is not plurality as such, but governance-relevant admissibility-bearing plurality: the claim that multiple standing-bearing interiors can inherit joint comparison, measurement, selection, or explanatory standing without explicit target gating or without collapsing into the same admissible interior. The result clarifies the structural pressure behind anthropic, landscape, multiverse, and modal parameter-space arguments. Such constructions may be used as explicit models if they supply their own target gating, transport licenses, compatibility constraints, witness bridges, and bounded observer-localization structure. They do not inherit those structures by multiplicity alone.

Orientation. This manuscript is the sixth paper in the constants-and-ensemble arc. The earlier papers established fixed-domain standing fixity, role classification of physical constants, renormalization as standing-preserving scale transport, a parameter mobility theorem, and the anthropic collapse theorem. The present paper does not repeat those results. It extracts their common consequence: ensemble multiplicity is admissible only when its embedding preserves standing, transport, localization, continuation loci, multiplicity discipline, and operator-domain discipline. If it does not, the ensemble is not an admissible standing-bearing comparison structure.

Contents

1	Introduction	3
1.1	Why this is not modal collapse	4
1.2	Embedding-dependency diagram	4
1.3	The ten-layer structure	4
1.4	Ensemble-collapse architecture	5
2	Scope and Non-Claims	6
2.1	What is not claimed	6
2.2	What is claimed	6
2.3	Terminology discipline	6
2.4	Three kinds of multiplicity	7
3	Minimality and Dependence Clarification	7
4	Fixed-Domain Admissibility Substrate	8
4.1	Ensemble objects	8
5	No Admissibility-Relevant Parameterization	9
6	No Competing Same-Domain Classifiers	9
7	No Admissibility-Preserving Extension	10
8	Cross-Domain Closure and Regime Translation Discipline	11
9	Scope Transport and Standing Transfer	12
10	No Primitive Ensemble Generators	13
10.1	Obstruction table	14
11	Transport Licensing and Compatibility	14
12	Continuation Loci and Multiplicity Constraints	15
13	Localization and Anti-Diffuse Persistence	17
14	Observer-Predicate Embedding Collapse	18
15	Embedding Normal Form	19
16	Ensemble Embedding Collapse Theorem	19
16.1	The collapse is finite	22
17	Extensional Plurality versus Admissible Plurality	23
18	Implications for Anthropic, Landscape, and Branch Ensembles	24
18.1	Anthropic ensembles	24
18.2	Landscape ensembles	24
18.3	Everettian branch structures	24

18.4 Modal parameter sweeps	24
19 Worked Examples	24
19.1 Fine-structure-constant scan	25
19.2 Cosmological constant anthropic bound	25
19.3 String-landscape-style EFT vacua	25
19.4 Everettian outcome branching	26
19.5 Generic modal worlds	26
20 No Metaphysical Overclaim	26
21 Objection Matrix	27
22 Conclusion	28
A Theorem Dependency Ledger	28
B Claim Index	29
C Short Proof Form	30

1 Introduction

Ensemble reasoning is widespread in foundational physics, cosmology, and philosophy of physics. Fine-tuning arguments compare alternative constant assignments. Anthropic arguments condition on observer-permitting members of a proposed domain of alternatives. Landscape models compare vacua. Everettian and modal discussions compare branches, outcomes, or possible worlds. In each case the proposed explanation depends on a plurality of alternatives that are distinct enough to carry explanatory contrast, but comparable enough to be treated as members of one standing-bearing ensemble.

That double demand is nontrivial. Distinct alternatives must be compared, measured, transported, classified, and sometimes conditioned on. They must share enough structure for constants, observers, records, evidence, and predicates to remain identifiable across the family. Yet they must differ enough for their differences to matter. The present paper asks whether such plurality can inherit standing-bearing ensemble status merely by being collected into a comparison regime.

The answer developed here is negative in a precise sense. The result is not that there is one metaphysical universe, one formal model, or one possible description. It is that admissibility-bearing plurality does not inherit standing-bearing authority from multiplicity alone. Once standing-bearing use is claimed, the family must pass the combined constraints of standing preservation, no-selector discipline, transport licensing, compatibility licensing, operator exhaustion, bounded localization, and admissible-interior uniqueness. A proposed ensemble either collapses extensionally into the same admissible interior, remains representational or bookkeeping plurality, or introduces structure that must be gated as a new target domain rather than treated as inherited standing.

The central theorem is therefore an embedding theorem, not an ontological thesis:

A standing-bearing ensemble is admissible only if its members can be jointly embedded into one standing-preserving comparison structure. Under the closure constraints proved below, every nontrivial embedding attempt either collapses as independent inherited

standing, becomes bookkeeping, is retyped as an explicitly gated target domain, or fails one of the required admissibility conditions.

This is the structural form of ensemble collapse.

1.1 Why this is not modal collapse

The theorem does not deny that different formal structures can be written down, that different physical models can be studied, or that counterfactual reasoning can be useful. It denies only that such plurality automatically carries admissibility-bearing standing. A formal alternative that lacks target gating, transport, witness compatibility, and bounded localization may still be a mathematical expression or exploratory model. It is not thereby a standing-bearing member of an admissible comparison ensemble.

This distinction is essential. The paper targets governance-relevant standing, not syntactic multiplicity. It is compatible with ordinary model building, effective theory analysis, counterfactual heuristics, and mathematical plurality. It constrains only the interpretive move that promotes a collection of alternatives into an admissible ensemble capable of explaining, measuring, selecting, or comparing standing-bearing physical structure.

1.2 Embedding-dependency diagram

A proposed standing-bearing ensemble must pass through several distinct layers before it can support explanation. Figure 1 summarizes the dependency order. The theorem below does not deny that a formal family may be written down at the first layer. It denies that such a family becomes standing-bearing until the later layers are admitted.

1.3 The ten-layer structure

The proof proceeds through ten layers:

- (i) no admissibility-relevant parameterization of the standing boundary;
- (ii) no competing same-domain admissibility classifiers;
- (iii) no admissibility-preserving extension that expands standing-bearing coverage on the original carrier;
- (iv) no cross-domain closure escape or hybrid-domain laundering;
- (v) no standing transfer, generator, carrier, repair, or selector as primitive ensemble machinery;
- (vi) no automatic cross-domain transport or transport compatibility;
- (vii) no trace-only continuation without a preserved continuation locus;
- (viii) no unbounded or locally indistinguishable multiplicity as standing-bearing structure;
- (ix) no diffuse standing-bearing persistence where bounded localization is required;
- (x) no governance-relevant multiplicity that survives all previous closures.

The ensemble embedding theorem follows by exhaustion. A proposed ensemble must use at least one of the routes these layers classify. Each route either preserves the same admissible interior, changes only presentation, changes domain, or fails admissibility.

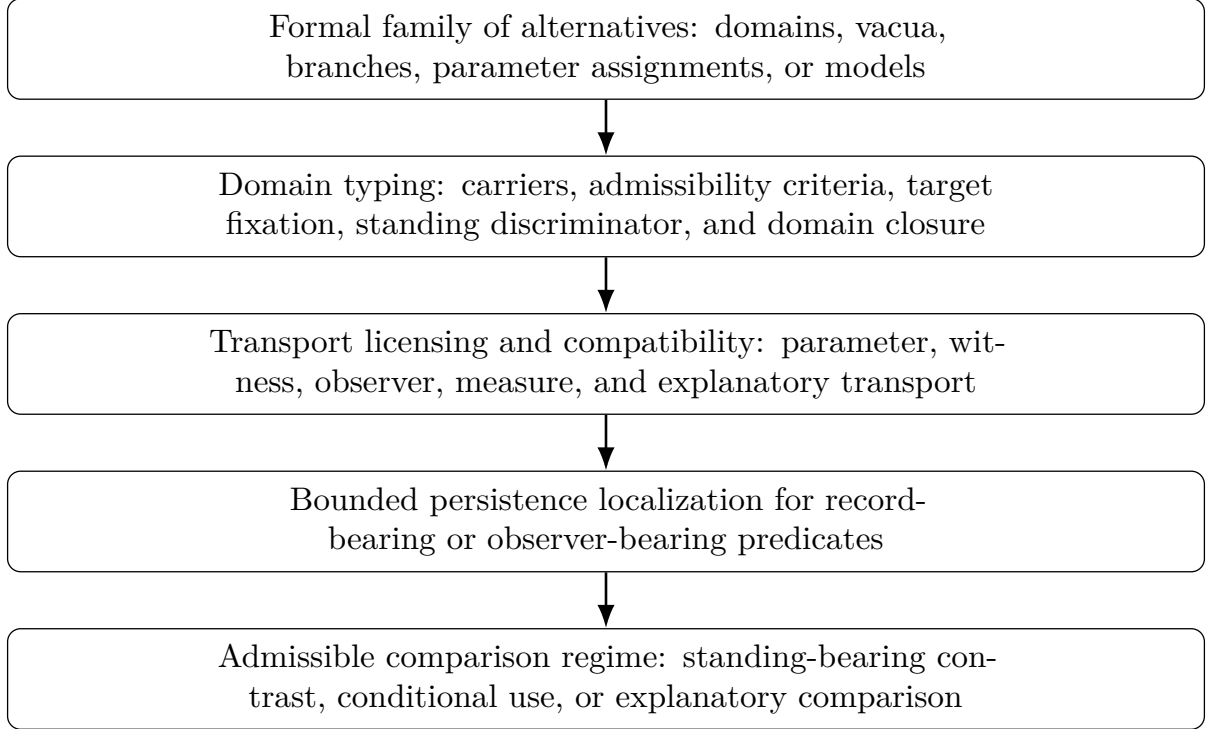


Figure 1: The ensemble embedding dependency stack. A formal family is not yet a standing-bearing ensemble. Standing-bearing ensemble use requires domain typing, transport and compatibility licensing, localization of persistence predicates where relevant, and a final admissible comparison regime.

1.4 Ensemble-collapse architecture

The collapse argument has a simple architecture. A multiplicity claim becomes relevant only when it is used as admissibility-bearing structure. Once used in that way, it requires selectors, transports, compatibility constraints, continuation loci, multiplicity witnesses, localization, or extension. Those routes are precisely the routes exhausted below.

Ensemble collapse architecture.

1. A multiplicity claim supplies many domains, vacua, branches, models, parameter assignments, or observer classes.
2. If the multiplicity is used only representationally, it remains formal or bookkeeping plurality.
3. If it is used governance-relevantly, it must supply admissibility-relevant differentiation.
4. Such differentiation requires selector/measure authority, licensed transport, compatibility, domain typing, continuation loci, bounded localization, or admissibility-preserving extension.
5. The closure tests below reduce every such route to extensional equivalence, bookkeeping plurality, fresh target gating, or inadmissibility.

2 Scope and Non-Claims

The result is intentionally narrow. It is not an unrestricted metaphysical thesis.

2.1 What is not claimed

The paper does not claim:

- that there is one material universe;
- that all possible worlds are logically identical;
- that modality is false;
- that formal model plurality is impossible;
- that all coordinate systems, representations, or descriptions are the same;
- that multiverse or landscape models are meaningless;
- that anthropic conditionalization is always illegitimate;
- that physics must abandon parameter scans, counterfactual modeling, or effective theory methods;
- that biological observers, material bodies, spacetime, or empirical systems are derived here.

2.2 What is claimed

The paper claims:

No inherited independently standing-bearing ensemble multiplicity is conferred by multiplicity alone. Any surviving multiplicity must be retyped as an explicitly gated target domain with its own transport, compatibility, witness, localization, and closure conditions. Within such a target domain, governance-relevant plurality is subject to the same admissible-interior uniqueness, no-selector, and bookkeeping-collapse constraints that govern any fixed standing domain.

The result applies relative to a fixed admissibility-governed continuation substrate. It does not globally quantify over unconstrained formal possibility spaces, and it does not deny formal plurality outside standing-bearing use.

This is a theorem about admissible comparison structure. It is not a theorem about substance, actuality, metaphysical existence, or empirical cosmology.

2.3 Terminology discipline

The word *interior* means a closure of standing-positive admissible content under admissibility-preserving operations. It does not mean spacetime interior or physical region. The word *embedding* means a standing-preserving inclusion of domains into a common comparison regime. It does not mean set-theoretic inclusion alone.

Definition 2.1 (Collapse). *In this paper, collapse means failure to survive as independent governance-relevant admissibility-bearing structure. A collapsed route may leave syntactic, formal, representational, or bookkeeping plurality intact. Collapse does not mean physical destruction, erasure of notation, denial of formal alternatives, or metaphysical elimination of plurality.*

2.4 Three kinds of multiplicity

The theorem distinguishes three cases that are often conflated.

Definition 2.2 (Formal multiplicity). *Formal multiplicity is the mere presence of many expressions, models, worlds-in-notation, parameter assignments, branches, histories, vacua, or descriptions. Formal multiplicity is permitted, but it is not standing-bearing by default.*

Definition 2.3 (Inherited ensemble multiplicity). *Inherited ensemble multiplicity is the claim that a family acquires standing-bearing comparison authority merely because many members are written, generated, counted, branched, or parameterized. This is the primary target of the theorem.*

Definition 2.4 (Explicitly gated target-domain multiplicity). *Explicitly gated target-domain multiplicity is multiplicity admitted only after the proposed comparison layer independently supplies domain typing, target gating, transport rules, compatibility constraints, witness preservation, and any measure or observer predicate it uses. Such multiplicity is not primitive inherited standing; it is a newly admitted target domain and is evaluated internally by the same closure conditions.*

Remark 2.5 (Survival condition for explicit ensemble theories). *The theorem does not ban a theory from defining an ensemble. It denies that ensemble standing is inherited from plurality alone. If an ensemble theory independently supplies the required gate, transport, compatibility, witness, localization, and measure structure, then the comparison is relocated into that target domain. Its standing is then the standing of that target domain, not a free inheritance from the original multiplicity.*

3 Minimality and Dependence Clarification

A skeptical reading might treat the main theorem as smuggling uniqueness into the setup. That is not the structure of the argument. The collapse result uses several independently stated closure axes, each with a distinct role. No single phrase such as “admissible”, “standing”, or “ensemble” is made to do all the work.

Definition 3.1 (Closure-axis dependence). *A closure-axis dependence is a load-bearing dependency without which at least one degeneracy route reopens. In this paper the relevant axes are: no admissibility-relevant boundary parameterization, no competing same-domain classifiers, no same-state repair, no primitive carriers or generators, no standing transfer, no admissibility-preserving total extension, licensed transport, licensed compatibility, continuation-locus preservation, multiplicity nondegeneracy, and bounded localization of standing-bearing persistence.*

Proposition 3.2 (No single-assumption uniqueness). *The Ensemble Embedding Collapse Theorem does not follow from any single closure axis alone. It follows from the joint exhaustion of the routes by which multiplicity could become governance-relevant.*

Proof. No boundary parameterization blocks indexed admissible alternatives, but by itself it does not analyze transport, localization, or extension. No competing classifier blocks same-domain classifier divergence, but by itself it does not block cross-domain scope shift. Operator exhaustion blocks generators and carriers, but by itself it does not provide transport compatibility. Licensed transport gives comparison rules, but by itself it does not guarantee compatibility among several transports. Bounded localization constrains persistence predicates, but by itself it does not classify measures or selectors. Multiplicity nondegeneracy blocks unbounded indistinguishable replication, but by itself it does not settle cross-domain translation. The main theorem requires the conjunction because ensemble multiplicity can attempt to enter through any of these routes. $\square$

Remark 3.3 (How to falsify the closure route). *The proof is conditional and route-sensitive. A counterexample must exhibit a standing-bearing ensemble embedding that avoids all listed routes: no boundary indexing, no auxiliary classifier, no repair/generator/carrier, no standing transfer, no scope change, no transport incompatibility, no unlicensed cross-domain overlap, no unbounded standing-indistinguishable replication, no trace-only continuation, and no failure of bounded localization. The theorem states that such a route is unavailable under the imported closure axes.*

4 Fixed-Domain Admissibility Substrate

The theorem begins from a fixed-domain substrate. This section records the minimal objects needed for the later embedding result.

Definition 4.1 (Fixed standing domain). *A fixed standing domain is a tuple*

$$\mathcal{D} = (X, \mathbf{A}, \mathbf{S}, \rightsquigarrow, \sim, \mathcal{W}),$$

where X is a class of evaluated states or constructional units, $\mathbf{A} : X \rightarrow \{0, 1\}$ is admissibility status, $\mathbf{S} : X \rightarrow \{0, 1\}$ is standing status, $x \rightsquigarrow y$ is licensed same-domain continuation, $x \sim y$ is lawful identity-preserving redescription, and $\mathcal{W}$ is the domain's witness ledger. Undefinedness is treated fail-closedly as non-instantiation of evaluation, not as a third standing value.

Definition 4.2 (Standing-bearing content). *A feature of $\mathcal{D}$ is standing-bearing if changing it alters admissible fixation, standing-bearing continuation, witness compatibility, identity-preserving transport, or coherent reuse in the fixed domain. A feature that changes only notation, presentation, coordinate description, or representational skin is not standing-bearing.*

Definition 4.3 (Admissible interior). *An admissible interior $\mathcal{I}(\mathcal{D})$ is the maximal closure of standing-positive admissible content under admissibility-preserving operations in $\mathcal{D}$. Two interiors are extensionally equivalent, written $\mathcal{I}_1 \equiv_{\text{Int}} \mathcal{I}_2$, when they induce the same standing-bearing classifications, continuation behavior, witness compatibility, and admissible redescription orbits.*

Definition 4.4 (Admissible distinguishability). *Two purported interiors are admissibly distinguishable if there exists a standing-bearing admissible witness, predicate, continuation, or transport relation that differentiates them inside the comparison regime. If no such witness exists, the distinction is bookkeeping or presentation-level only.*

Remark 4.5 (Why interiors, not worlds). *The theorem speaks of interiors rather than worlds. A world-language version would invite metaphysical readings that are not needed. The relevant object is the admissible standing-bearing closure generated by a fixed domain, not a maximal possible reality in a modal ontology.*

4.1 Ensemble objects

Definition 4.6 (Standing-bearing ensemble). *A standing-bearing ensemble is a family*

$$\mathcal{E} = \{\mathcal{D}_i\}_{i \in I}$$

with a proposed common comparison regime $\mathcal{C}_{\mathcal{E}}$ such that each member is intended to carry standing-bearing content and the family is intended to support at least one admissibility-relevant comparison, selection, measure, transport, observer predicate, or explanatory contrast.

Definition 4.7 (Ensemble embedding). *An ensemble embedding of $\mathcal{E}$ is a standing-preserving construction*

$$\iota_i : \mathcal{D}_i \longrightarrow \mathcal{C}_{\mathcal{E}}$$

for every $i \in I$, together with comparison operations in $\mathcal{C}_{\mathcal{E}}$ that preserve target fixation, witness compatibility, admissibility status, standing status, and licensed transport wherever the ensemble makes standing-bearing claims.

Definition 4.8 (Independent ensemble multiplicity). *An ensemble has independent admissibility-bearing multiplicity if at least two embedded members are admissibly distinguishable and the distinction is used as standing-bearing explanatory, evidential, selector, measure, or observer-predicate content.*

The main theorem denies survival of independent admissibility-bearing ensemble multiplicity. It does not deny representation-level or bookkeeping plurality.

5 No Admissibility-Relevant Parameterization

The first pressure on ensembles comes from the AMetric boundary. If standing at the admissibility interface is binary and non-eventful, then admissibility-relevant parameterization cannot be smuggled into the boundary as a selector.

Definition 5.1 (Admissibility-relevant parameterization). *An admissibility-relevant parameterization is an index, coordinate, ordering, metric, measure, ranking, or selector that distinguishes multiple positive admissibility statuses in a way that affects standing-bearing eligibility.*

Theorem 5.2 (No admissibility-relevant parameterization). *Within a fixed standing domain, there is no admissibility-relevant parameterization of boundary crossing, including discrete indexing of multiple admissible alternatives.*

Proof. Standing at the boundary is bivalent on the evaluated fragment: admissible or not admissible. If a parameter $i \in I$ distinguishes admissible positives in a way that affects eligibility, then the boundary has more than two standing-relevant statuses: negative, positive-of-type- i , and positive-of-type- j for $i \neq j$. That contradicts bivalence. If the index is treated as chosen by a selector, then selection is a boundary event or operator; but the boundary is non-eventful and operators are defined only after admissibility. If the index is inert, it is not admissibility-relevant. Hence no admissibility-relevant parameterization exists. $\square$

Corollary 5.3 (No ensemble indexing by boundary authority). *A family $\{\mathcal{D}_i\}_{i \in I}$ does not become an admissible standing-bearing ensemble merely by being indexed. The index either is bookkeeping, requires a selector, or changes domain.*

Remark 5.4 (Continuous and discrete cases). *The same result applies to continuous ranges and discrete plural alternatives. A continuous parameter space of possible constants and a discrete list of vacua both require admissibility-relevant differentiation if used as standing-bearing ensemble structure. The obstruction is not cardinality. It is boundary authority.*

6 No Competing Same-Domain Classifiers

The second pressure comes from classifier uniqueness. An ensemble cannot carry multiple incompatible standing classifiers while remaining one fixed comparison regime.

Definition 6.1 (Same-domain classifier). *A same-domain classifier is a map $G : X \rightarrow \{0, 1\}$ claimed to govern admissible fixation, standing-bearing reuse, or coherent continuation on the same fixed domain.*

Theorem 6.2 (No competing same-domain classifiers). *Let $G, G' : X \rightarrow \{0, 1\}$ be lawful same-domain classifiers claimed to govern admissibility or standing-bearing reuse on the same fixed domain. If G and G' are governance-relevant, then $G = G'$ pointwise. Any disagreement requires a forbidden higher-order selector or produces indeterminate standing.*

Proof. Assume there is $x \in X$ with $G(x) \neq G'(x)$. If both classifications are allowed to govern, then the same evaluated state receives conflicting same-domain standing status, contradicting bivalence. If one classification is chosen as authoritative, then the system requires a higher-order gate chooser to determine which classifier governs. Such a chooser is an admissibility-relevant selector and is blocked by the AMetric no-selector boundary. Hence no disagreement is admissible. $\square$

Theorem 6.3 (Governance-role collapse). *Any same-domain distinction that preserves pointwise standing status but purports to alter governance role is bookkeeping unless it changes evaluated-state classification or coherent legal trajectory structure. If it changes either, it introduces a second classifier, repair, generator, carrier, or standing modification and collapses.*

Proof. Same-domain governance-relevant behavior is realized at two loci: state classification and coherent continuation. If a distinction changes neither, it has no governance effect and is bookkeeping. If it changes classification, it conflicts with the unique same-domain classifier. If it changes coherent continuation while preserving pointwise standing, it must import an auxiliary rule for retained trajectories; that rule is either a second classifier, a same-state repair, a generator, or a standing modifier. Each is inadmissible under the fixed-domain closure conditions. $\square$

Corollary 6.4 (Same-domain ensemble collapse). *An ensemble whose members are embedded as same-domain standing structures either collapses extensionally into the same admissible interior or differs only by presentation.*

7 No Admissibility-Preserving Extension

A natural escape route is to claim that an ensemble is not a competing classifier but an extension. This section blocks that route for admissibility-bearing extension on the original carrier.

Definition 7.1 (Admissibility-preserving extension). *An extension $\mathcal{D} \rightsquigarrow \mathcal{D}^*$ is admissibility-preserving on the original carrier when the standing-bearing classifications, operator domains, continuation relations, and witness meanings of $\mathcal{D}$ are preserved inside $\mathcal{D}^*$.*

Definition 7.2 (Standing-extending operator attempt). *A standing-extending operator attempt is an operator family that purports to make previously undefined or inadmissible standing-bearing acts admissible, or to expand admissible comparison coverage, without resetting domain identity and regating the target domain.*

Theorem 7.3 (Extension collapse). *Every admissibility-preserving extension of a non-degenerate standing domain on its original carrier falls into exactly one of three classes:*

- (a) *definitional or presentation-level extension;*
- (b) *non-conservative scope change requiring new target gating;*

(c) inadmissible extension by repair, generator, carrier, selector, or scope transport.

No fourth extension class expands standing-bearing coverage while preserving original domain identity.

Proof. If the extension preserves all original standing-bearing consequences, operator domains, and witness meanings, then it is definitional or presentation-level on the original carrier. If it changes standing-bearing consequences, then either it changes the scope and therefore requires new gating, or it attempts to alter standing on the same carrier. The latter is a repair, generator, carrier, selector, or scope transport. The finite exhaustion of standing-extending operators blocks each case. Since expanding standing-bearing coverage must either change nothing, change domain, or change standing on the original domain, the listed cases are exhaustive. $\square$

Corollary 7.4 (Ensemble extension corollary). *A purported ensemble cannot enlarge admissibility-bearing comparison structure while remaining a same-domain admissibility-preserving extension. It is either definitional, a new domain, or inadmissible.*

Remark 7.5 (Exploratory models). *This result does not prohibit exploratory extension, model-building, or formal augmentation. It constrains only the interpretive claim that such augmentation inherits standing-bearing authority over the original domain without target gating or scope change.*

8 Cross-Domain Closure and Regime Translation Discipline

The preceding extension result is same-domain. Ensemble arguments often attempt a stronger move: they pass from one admitted domain to a hybrid or cross-domain comparison layer, then use that layer to repair or compare structures that were not comparable in either source domain alone. The corpus supplies a separate closure discipline for that move.

The cross-domain closure certificates are used here as closure summaries, not as substitutes for proof inside the present manuscript. The operative proof obligations below are discharged by explicit target gating, overlap non-divergence, regime-translation discipline, transport licensing, compatibility licensing, scope discipline, and witness preservation.

Definition 8.1 (Cross-domain escape). *A cross-domain escape is an attempted ensemble embedding that repairs failure, supplies missing standing, introduces comparability, or licenses selection by moving between mathematical, physical, modal, informational, computational, or hybrid regimes without an independently admitted target gate.*

Theorem 8.2 (Cross-domain no-escape principle). *A failed standing-bearing comparison in one domain cannot be repaired by transfer into another domain unless the target comparison layer is independently gated, typed, and closed. A hybrid or cross-domain layer that lacks such gating is not an admissible ensemble embedding.*

Proof. If the target layer has no independent gate, it has no authority to classify standing-bearing content. If it borrows authority from the source, it is standing transfer. If it changes the envelope to legalize the comparison, it is scope transport. If it introduces a new selector, measure, or primitive to arbitrate the comparison, it is carrier or selector structure. If it is independently gated and closed, then it is a new admitted domain, not inherited cross-domain standing. These possibilities exhaust the cross-domain escape route. $\square$

Lemma 8.3 (Overlap non-divergence). *Let two admissible representational regimes overlap on a normalized object-class and claim standing-bearing comparison on that overlap. They cannot disagree on the overlap without either introducing an arbitration primitive, privileging one representation, or changing the target domain.*

Proof. If both incompatible overlap claims are admitted, standing conservation fails on the overlap. If one claim is privileged, a selector or arbitration rule has been introduced. If the conflict is removed by changing the object-class or target, the comparison has changed domain. Thus admissible overlap comparison is non-divergent once normalization and target identity are fixed. $\square$

Definition 8.4 (Admissible regime translation). *An admissible regime translation is a cross-domain lift that preserves admissible morphisms, fixed target identity, factorization coherence, non-fragmentation, and standing-bearing witness content. It must not introduce irreversible admissible embeddings, irreversible admissible quotients, infinite strict-loss refinement cascades, or incompatible decomposition paths.*

Proposition 8.5 (Regime translation discipline). *A cross-domain embedding used for standing-bearing ensemble comparison is admissible only if its translation layer preserves admissible morphisms, admits path-independent factorization up to admissible isomorphism, avoids irreconcilable fragments, and does not rely on irreversible embeddings or quotients as standing-bearing authority.*

Proof. A translation layer is being used to carry standing from one regime into another. If admissible morphisms are not preserved, the translation fails as transport. If factorization is not coherent, the same comparison can decompose into incompatible standing claims. If fragments are irreconcilable, no single comparison domain is fixed. If irreversible embeddings or quotients supply the standing-bearing step, the purported comparison changes the admissible carrier without reversible witness structure. Each case is either transport failure, scope change, or fresh target gating. $\square$

Corollary 8.6 (No hybrid-domain laundering). *Hybrid domains do not inherit admissibility from their components. A mathematical-physical, computational-physical, modal-physical, or informational-physical hybrid must be admitted as its own target domain before it can carry ensemble standing.*

9 Scope Transport and Standing Transfer

The previous section blocks nontrivial admissibility-preserving extensions. A related failure occurs when an ensemble changes the operative domain after the fact while retaining the standing authority of the original domain. This is scope transport. Another failure occurs when standing in one domain is treated as standing in another without independent target gating. This is standing transfer.

Definition 9.1 (Scope transport). *Scope transport is any attempt to alter, expand, narrow, or reframe the operative domain after the relevant standing-bearing claim has been fixed, while treating the altered domain as if it inherited the original standing authority.*

Definition 9.2 (Standing transfer). *Standing transfer is any claim that standing in a target domain D_2 is licensed by standing in a source domain D_1 , without independent target gating and domain-of-application closure in D_2 .*

Theorem 9.3 (No illicit scope transport). *A purported ensemble embedding that changes the operative domain to legalize an otherwise inadmissible comparison is not an admissible same-regime embedding. It is a scope change requiring fresh gating.*

Proof. The scope of a standing-bearing construction is determined by the active envelope. If an ensemble comparison fails under that envelope, changing the envelope while retaining the original standing claim is not clarification but retyping. Retyping may define a new target domain, but it does not preserve standing in the original one. Thus the move is either fresh gating on a new domain or illicit scope transport. $\square$

Theorem 9.4 (No standing transfer without target gating). *Let D_1 and D_2 be distinct standing-governed domains. Standing in D_1 does not license standing in D_2 unless D_2 independently instantiates its own admissibility classifier, target fixation, and domain-of-application closure.*

Proof. If D_1 's classifier is used to classify D_2 , the source envelope has been substituted for the target envelope, which is scope transport. If a derived similarity, performance, measure, or analogy licenses transfer, that derived predicate functions as a carrier. If D_2 independently gates the relevant content, then the standing arises in D_2 and is not inherited from D_1 . These exhaust the routes. $\square$

Corollary 9.5 (No source-domain authority for ensembles). *No member of an ensemble confers standing on the ensemble as a whole. The ensemble must be gated as its own comparison domain.*

Remark 9.6 (Why this matters for multiverse and landscape claims). *A formal model may contain many members. But the standing of a particular member, calculation, branch, or vacuum does not automatically become standing of the ensemble-level comparison. The comparison domain must itself be admitted.*

10 No Primitive Ensemble Generators

An ensemble may be presented as generated by a rule: a probability distribution, a landscape construction, a selection principle, a typicality measure, a canonical representative procedure, a branch-weighting scheme, or a modal closure. Such a generator must be classified.

Definition 10.1 (Primitive ensemble generator). *A primitive ensemble generator is a rule Γ that purports to produce an admissible standing-bearing ensemble from a source domain without independent target gating, transport licensing, compatibility constraints, and witness preservation for the generated members.*

Theorem 10.2 (Ensemble generation exhaustion). *Every primitive ensemble generator instantiates at least one of the following obstruction classes:*

- (E1) *boundary selector or admissibility-relevant index;*
- (E2) *repair or generator of standing from a prior inadmissible or undefined act;*
- (E3) *carrier laundering by probability, typicality, measure, symmetry, explanation, or structural invariant;*
- (E4) *scope transport or domain retyping.*

Each class is inadmissible as primitive ensemble generation.

Proof. A primitive generator must do one of four things. It may select a member or index at the boundary, which is a boundary selector. It may turn an undefined or inadmissible construction into a standing-bearing member, which is repair or generation. It may authorize membership by a derived criterion such as measure, typicality, symmetry, or explanatory power, which is carrier laundering.

Or it may widen or retype the domain so the generated members count as comparable, which is scope transport. These cases exhaust primitive generation. Each is blocked by the standing-gate and operator-exhaustion results. $\square$

Corollary 10.3 (No primitive measure ensemble). *A probability distribution, typicality rule, branch weight, vacuum count, or measure dominance relation cannot generate admissible ensemble standing. Such structures may operate only after an ensemble has been independently admitted.*

Corollary 10.4 (No canonical representative ensemble). *A canonical representative or quotient selection procedure cannot by itself produce standing-bearing ensemble membership. If it selects, it requires admitted selector authority; if it does not select, it is bookkeeping.*

10.1 Obstruction table

The table is not a sociological critique of how these tools are used in physics, mathematics, or cosmology. It classifies only their attempted use as primitive standing-authorizing ensemble machinery.

Proposed generator	Standing-bearing role claimed	Forced obstruction
Parameter-space interval	Treats coordinates as admissible alternatives	Boundary selector or scope transport
Landscape measure	Weights domains or vacua for explanation	Carrier unless independently gated
Typical observer rule	Converts observer count or typicality into authority	Carrier laundering
Everett branch counting	Treats outcome multiplicity as law/constant multiplicity	Scope transport
Canonical representative	Selects a privileged member of an equivalence class	Selector
Model-theoretic existence	Promotes formal realization into standing-bearing membership	Generator or scope change
Universal-property construction	Converts defining property into admissible existence	Generator or retyping
Observer averaging	Converts many observer-like records into standing authority	Carrier or generator
Infinite ensemble completion	Treats unbounded completion as admissible comparison space	Totalization or scope transport
Ensemble repair	Uses the ensemble to rehabilitate failed member standing	Repair/generator
Cross-domain hybridization	Moves a failed comparison into a hybrid regime	Scope transport unless independently gated
Explanatory unification	Makes unification authorize standing	Carrier or repair

11 Transport Licensing and Compatibility

Even if an ensemble does not rely on primitive generation, it must still compare its members. Comparison requires transport. Transport is not primitive.

Definition 11.1 (Licensed transport). *A licensed transport schema is a triple*

$$(\mathcal{L}, D, \mathcal{T}_{\mathcal{L}}),$$

where $\mathcal{L}$ is an explicit ex ante license, D is a fixed identification domain, and $\mathcal{T}_{\mathcal{L}}$ is a standing-preserving transport rule operating only on elements of D .

Theorem 11.2 (No automatic cross-domain transport). *Transport between standing-bearing domains is not inherited by formal juxtaposition, common notation, metric articulation, or ensemble membership. It exists only when explicitly licensed over a fixed identification domain and standing is preserved by the transport rule.*

Proof. A transport rule maps relational content between relata. If no license and domain are specified, the mapping has no admissible scope. If the rule is applied outside its domain, it is scope transport. If it fails to preserve standing, it is not admissible transport. Hence transport is not automatic. $\square$

Definition 11.3 (Transport compatibility). *A family of transports is compatible when an explicitly licensed higher-order constraint authorizes their joint use over a fixed domain and all applications preserve standing globally.*

Theorem 11.4 (No automatic transport compatibility). *Distinct transports are not mutually compatible by default. Compatibility among parameter transport, observer transport, measure transport, branch transport, evidence transport, and explanatory transport requires explicit higher-order licensing.*

Proof. Compatibility is a relation among transport rules. A transport license authorizes a rule; it does not automatically authorize relations among multiple rules. Such relations impose additional constraints and therefore require an additional license. Without that license, the transports may each be locally defined while failing joint standing preservation. $\square$

Theorem 11.5 (Transport-holonomy obstruction for continuous ensembles). *If a continuous parameter family in an ensemble induces standing-distinguishable holonomy under closed admissible transport, then the family is inadmissible as a continuous standing-bearing comparison space unless it quotients into transport-invariant admissible classes.*

Proof. A closed admissible transport must return standing-bearing content to standing-identical form. If a loop maps a continuous parameter λ to $\lambda' \neq \lambda$ in a standing-distinguishable way, then standing is not preserved. The only admissible alternatives are elimination as redescription artifact, quotienting into transport-invariant classes, or classification as a transport-neutral envelope input. A genuinely continuous standing-distinguishable holonomy remains inadmissible. $\square$

Corollary 11.6 (No free ensemble comparability). *A standing-bearing ensemble requiring parameter, observer, measure, or evidence comparison must supply transport and compatibility licenses. Without them, ensemble comparability is undefined or inadmissible.*

12 Continuation Loci and Multiplicity Constraints

Transport licensing is not enough. Ensemble members must also preserve what is being continued. A trace, label, branch number, vacuum index, or parameter coordinate can record formal succession without preserving a standing-bearing continuation locus.

Definition 12.1 (Continuation locus). *A continuation locus is the minimal admissibility-bearing identity locus preserved by a same-domain continuation claim. It is not a substance, object, semantic denotatum, or hidden bearer. It is the quotient-level preservation behavior by which a regime distinguishes continuation from trace succession, replacement, relabeling, or scope change.*

Theorem 12.2 (Continuation-locus induction). *Every non-degenerate continuation-bearing governance regime induces a continuation-locus equivalence class for each witnessing load-bearing continuation role:*

$$\text{NonDegCont}(D) \longrightarrow [L_D]_{\equiv \text{Loc}}.$$

Proof. A non-degenerate continuation-bearing claim is not trace-only and not inert bookkeeping. It must preserve some same-domain identity through lawful passage for later reuse, commitment, construction, exclusion, or counterexample burden. The preserved role is the continuation locus. Lawful redescription identifies presentations that preserve the same continuation eligibility, identity, transport, quotient behavior, standing-bearing reuse, and fail-closed structure. Thus each witnessing role induces a quotient class. $\square$

Corollary 12.3 (Trace insufficiency for ensembles). *A sequence of traces, labels, branch indices, vacua, or parameter values does not by itself supply standing-bearing continuation across an ensemble. If it preserves no continuation locus, it is trace-only or bookkeeping; if it preserves one, the locus must be included in the target governance structure.*

Definition 12.4 (Restriction footprint). *The restriction footprint of an admissible object is the family of admissible equivalence classes obtained from its admissible proper restrictions. Two objects have identical footprints when every admissible proper restriction yields equivalent standing-bearing content.*

Proposition 12.5 (Conditional footprint multiplicity strengthening). *Conditional on the SND+ strengthening, an infinite family of globally inequivalent standing-bearing admissible objects with identical admissible proper-restriction footprints is not available as standing-bearing ensemble multiplicity unless an additional admissible witness distinguishes the global alternatives.*

Proof. SND+ is treated here as conditional support rather than as a load-bearing premise of the main theorem. Under that strengthening, if all proper restrictions are admissibly equivalent, no admissible local restriction witnesses the claimed difference. The distinction is then maintained only at full scope. Without an admissible witness anchoring that full-scope inequivalence, the family carries nonlocal standing instability: identical admissible inputs are being used to support multiple inequivalent standing-bearing outcomes. The alternatives are quotient collapse, fresh witnessing structure, or loss of standing-bearing multiplicity. $\square$

Remark 12.6 (Dependency discipline for SND+). *The main theorem below does not require SND+ as an established closure result. The theorem uses only the weaker obligation that footprint-based multiplicity must be witnessed, quotient-stabilized, retyped, or treated as bookkeeping. SND+ records a stronger conditional consequence if the additional nondegenerate-footprint principle is adopted.*

Theorem 12.7 (No unbounded standing-indistinguishable replication). *Unbounded replication of structurally identical standing-indistinguishable sectors is not an admissible source of ensemble multiplicity. Such replication is either eliminable redundancy, finite non-degenerate envelope input, or transport-incoherent duplication.*

Proof. If replicated sectors are fully symmetric under admissible redescription, they carry no standing-distinguishable content and are eliminable. If a global label distinguishes them, that label must be standing-preserving and transport-neutral; otherwise it is illicit selector structure. Unbounded replication admits permutation transports whose quotient does not close without adding global structure. Hence any surviving multiplicity must be finite, non-redundant, and either forced by non-degeneracy or certified as an irreducible discrete envelope input. $\square$

Corollary 12.8 (Multiplicity-minimality pressure). *A standing-bearing ensemble cannot appeal to unbounded indistinguishable replication as explanatory depth. Multiplicity must be locally witnessed, transport-closed, finite or quotient-stabilized, and independently admitted.*

13 Localization and Anti-Diffuse Persistence

Ensemble embedding also requires observer-like or record-like predicates to persist across the proposed alternatives. The localization results show why such predicates cannot remain globally diffuse under standing-bearing persistence.

Localization clarification. The theorem concerns bounded standing-bearing persistence structure. It is not a claim about biological embodiment, physical organismhood, consciousness, empirical spacetime localization, or metaphysical individuality. Boundedness is role-structural: it marks the locus through which perturbation, repair, identity retention, and standing consequence are jointly evaluated.

Definition 13.1 (Realized persistence regime). *A realized persistence regime is a standing-bearing continuation structure in which consequence retention, identity memory, transport behavior, quotient behavior, failure behavior, and irreversibility profile are preserved under lawful same-domain redescription.*

Definition 13.2 (Bounded standing-localization equivalence). *A bounded standing-localization equivalence class $[B_D]_{\equiv \text{Bound}}$ is the equivalence class of bounded role-structural loci preserving boundary role, identity-memory profile, retained repair semantics, perturbation burden, standing margin, transport behavior, quotient behavior, and fail-closed status under lawful same-domain redescription.*

Theorem 13.3 (Standing-repair localization). *If a fixed domain supports realized standing-bearing persistence with lawful repair semantics, then standing cannot remain global, diffuse, trace-level, or unlocalized. It induces a bounded standing-localization equivalence class:*

$$\text{RealPers}_D \wedge \text{RepairSem}_D \longrightarrow [B_D]_{\equiv \text{Bound}}.$$

Proof. Realized persistence supplies standing-bearing consequence retention. Repair semantics distinguishes degradation, correction, retained identity, bounded repair locality, continued standing, and failure. These distinctions require a bounded identity locus; otherwise repair is global reindexing, envelope import, standing transfer, same-act repair, or bookkeeping. Global and diffuse standing fail to factor repair through such a bounded locus. Hence lawful repair semantics induces bounded standing localization. $\square$

Corollary 13.4 (No diffuse standing-bearing ensemble predicate). *A standing-bearing predicate used across an ensemble cannot remain globally diffuse if it must preserve realized persistence under perturbation and lawful repair. It must localize into bounded standing-localization structure or fail as a standing-bearing predicate.*

Localization scope clarification. The localization theorem concerns bounded standing-bearing persistence structure. It is not a claim about biological embodiment, physical organismhood, consciousness, material individuality, or empirical observerhood. In this paper it is used only to constrain globally diffuse observer, record, or persistence predicates when those predicates are asked to carry standing-bearing ensemble comparison.

14 Observer-Predicate Embedding Collapse

Anthropic and multiverse ensembles often use observer predicates, life-permitting predicates, record predicates, or measurement predicates. The present section treats such predicates as standing-bearing only when they support admissible evidence, conditioning, or explanatory comparison.

Definition 14.1 (Standing-bearing observer predicate). *An observer predicate Obs_D is standing-bearing on a domain D when it is used to classify, condition, select, measure, or explain admissible alternatives by reference to record-bearing or observer-like persistence. It is not required to be biological; it is required to be transportable if used across domains.*

Definition 14.2 (Observer-predicate embedding). *An observer-predicate embedding for an ensemble $\mathcal{E} = \{D_i\}_{i \in I}$ is a family of predicates Obs_i and transports into a common comparison regime such that observer-bearing status is preserved, comparable, and standing-bearing across the embedded members.*

Theorem 14.3 (Observer-predicate embedding collapse). *A standing-bearing observer predicate cannot be globally embedded across independently admissible domains without one of the following:*

- (a) *bounded standing-localization structure transported under explicit license;*
- (b) *extensional collapse into the same admissible interior;*
- (c) *bookkeeping-level predicate similarity;*
- (d) *illicit standing transfer;*
- (e) *scope drift or target retargeting;*
- (f) *selector, carrier, or measure injection;*
- (g) *transport or compatibility failure.*

No independent globally diffuse observer-predicate embedding remains.

Proof. If the predicate is standing-bearing, it must preserve record-like or observer-like persistence under comparison. By standing-repair localization, standing-bearing persistence with repair semantics localizes into bounded structure; global diffuse persistence is not admissible. If bounded structure is supplied and transported under license, the embedding is explicitly gated. If the transported structure preserves all standing-bearing roles, it collapses extensionally into the same admissible interior. If it changes only vocabulary, it is bookkeeping. If it supplies standing without target gating, it is standing transfer. If it changes the comparison envelope, it is scope drift. If it relies on typicality, measure, or observer count as primitive authority, it is a carrier. If transports are not licensed and compatible, comparison fails. The listed cases exhaust the possibilities. $\square$

Corollary 14.4 (Anthropic observer-class corollary). *An anthropic “observer-permitting” class is not automatically a standing-bearing subset of an ensemble. It becomes such only when observer-predicate transport, bounded localization, witness compatibility, and measure or weighting structure are independently admitted.*

15 Embedding Normal Form

The preceding sections isolate the obstruction families. Before stating the main theorem, it is useful to put the admissible case into normal form.

Definition 15.1 (Embedding normal form). *A proposed ensemble embedding is in normal form when it is presented as a tuple*

$$\mathfrak{N}(\mathcal{E}) = (\{\mathcal{D}_i\}_{i \in I}, \mathcal{C}_{\mathcal{E}}, \iota_i, \mathcal{T}, \text{Comp}, \text{Loc}, \text{Pred}, \text{Sel}, \text{Foot}),$$

where $\mathcal{C}_{\mathcal{E}}$ is the common comparison regime, ι_i are member embeddings, $\mathcal{T}$ is the family of licensed transports, **Comp** is the compatibility structure on composed transports, **Loc** records continuation-locus and bounded-localization structure where persistence is used, **Pred** contains standing-bearing observer, record, or persistence predicates, **Sel** contains any measure, selector, or typicality rule used for standing-bearing claims, and **Foot** records restriction-footprint or multiplicity data when the ensemble relies on replicated or locally indistinguishable members.

Proposition 15.2 (Normal-form reduction). *Every standing-bearing ensemble claim can be reduced to an embedding normal form, possibly with some components missing. If any required component is missing while the ensemble still makes a standing-bearing claim, the claim falls into one of the obstruction classes already identified.*

Proof. An ensemble claim must specify what its members are, how they are included in a common comparison regime, how standing-bearing content is transported, how multiple transports remain compatible, which continuation loci or bounded-localization structures preserve persistence predicates, how observer or record predicates are compared if used, how restriction-footprint or replication claims are witnessed if multiplicity is invoked, and how any measure or selector operates if probabilistic or typicality claims are made. These are precisely the components of the normal form. If a component is not supplied but the ensemble still uses it, the missing structure is being provided by boundary selection, standing transfer, carrier authority, scope transport, or unlicensed transport. Those are the obstruction classes already proved. $\square$

Theorem 15.3 (Admissible embedding normal-form theorem). *A normal-form ensemble embedding is admissible only if all standing-bearing components in $\mathfrak{N}(\mathcal{E})$ are independently typed, licensed, and compatibility-preserving inside $\mathcal{C}_{\mathcal{E}}$. If this condition holds, the embedding is either an explicit newly gated domain or a presentation of the same admissible interior. If it fails, the embedding is not standing-bearing.*

Proof. Typing is required for the members and comparison regime to be well-formed. Licensing is required for transport. Compatibility is required for composed transports to be jointly admissible. Predicate localization is required for observer or record-bearing persistence. Selector and measure structures must be downstream of admission. When all of these are supplied, the ensemble is no longer primitive: it is an explicit domain with its own gate and closure conditions. If its members preserve the same standing-bearing content, it presents the same admissible interior. If a component fails, the ensemble cannot carry the standing-bearing comparison it claims. $\square$

16 Ensemble Embedding Collapse Theorem

We now combine the preceding results. The theorem applies only relative to a fixed admissibility-governed continuation substrate; it does not quantify over unconstrained formal possibility spaces or deny formal plurality outside standing-bearing use.

Lemma 16.1 (Route exhaustion for ensemble multiplicity). *Any governance-relevant ensemble distinction must alter at least one of the following: domain typing, admissibility classification, target fixation, witness compatibility, transport, transport compatibility, quotient identity, continuation-locus preservation, bounded persistence localization, multiplicity footprint, replication structure, selector or measure authority, or operator-domain extension. These components exhaust the ways a formal family can be promoted into a standing-bearing comparison ensemble.*

Proof. A formal family becomes governance-relevant only by changing what is admitted, what is fixed as the target, what witnesses are comparable, what transports across members, what quotient identity is preserved, what continuation or persistence predicate survives, how multiplicity is distinguished, or what rule selects, measures, repairs, generates, or extends the family. If none of these components changes, the family is formal or bookkeeping plurality. If at least one changes, it is one of the listed components. The preceding sections classify each component: same-domain classifier structure, extension, cross-domain translation, scope and standing transfer, primitive generators, transport and compatibility, continuation loci, multiplicity constraints, and bounded localization. Hence no further governance-relevant route remains. $\square$

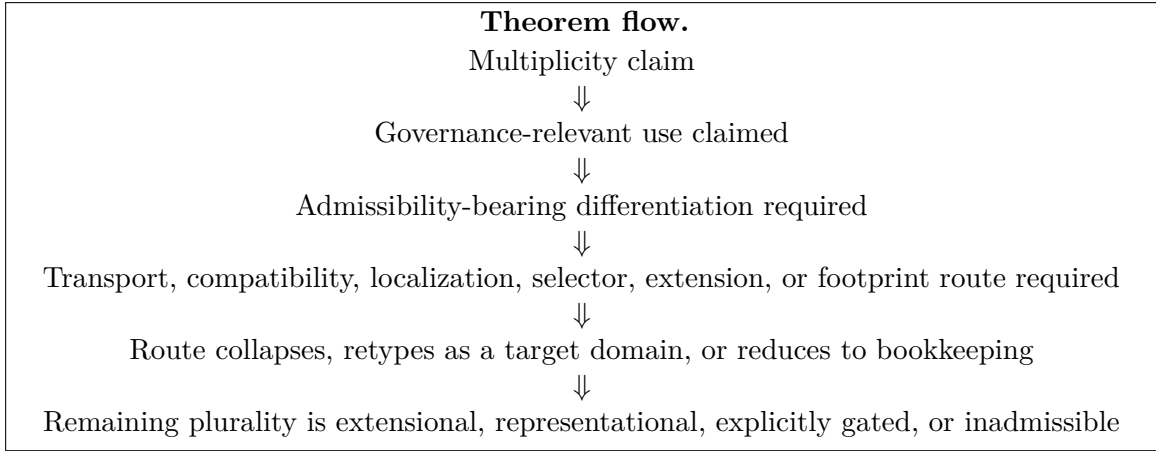


Figure 2: Logical flow of the main theorem. The result does not remove formal plurality. It denies only inherited governance-relevant ensemble standing unless the required routes are independently gated and then evaluated inside their target domain.

Definition 16.2 (Admissible ensemble embedding). *An admissible ensemble embedding is an ensemble embedding satisfying all of the following:*

- (a) *each member has independent target gating and domain-of-application closure;*
- (b) *all standing-bearing comparisons are transported under explicit license;*
- (c) *all composed transports are compatibility-licensed;*
- (d) *all measures, selectors, and typicality rules operate only inside an already admitted ensemble;*
- (e) *standing-bearing observer, record, or persistence predicates are supported by continuation-locus structure and bounded localization where relevant;*
- (f) *restriction-footprint or replication multiplicity is either locally witnessed, quotient-stabilized, finite envelope input, or bookkeeping;*

- (g) *no original-domain standing is repaired, generated, transferred, or authorized by a primitive carrier.*

Theorem 16.3 (Ensemble Embedding Collapse Theorem). *Let $\mathcal{E} = \{\mathcal{D}_i\}_{i \in I}$ be a purported ensemble of standing-bearing admissibility domains. If $\mathcal{E}$ is presented as an admissible comparison regime, then every attempt to treat its multiplicity as inherited independent standing resolves into at least one of the following outcomes:*

- (i) *the members collapse extensionally into the same admissible interior;*
- (ii) *the differences are bookkeeping or presentation-level only;*
- (iii) *the construction requires illicit scope transport or retargeting;*
- (iv) *the construction imports standing without independent target gating and target domain closure;*
- (v) *the construction requires admissibility-relevant parameterization or boundary selection;*
- (vi) *the construction relies on a primitive selector, carrier, generator, repair, typicality rule, or measure;*
- (vii) *the construction invokes cross-domain or hybrid-domain standing without an independently gated translation layer;*
- (viii) *the construction fails transport closure, transport compatibility, overlap non-divergence, or regime-translation discipline;*
- (ix) *the construction relies on unbounded standing-indistinguishable replication, or on unsupported footprint multiplicity under the optional SND+ strengthening;*
- (x) *the construction lacks continuation-locus structure for persistence claims;*
- (xi) *the construction fails bounded standing-localization for observer, record, or persistence predicates.*

Consequently, multiplicity alone never confers inherited independent standing-bearing authority. Any surviving standing-bearing multiplicity must be retyped as an explicitly gated target domain, where its governance-relevant plurality is subject to the same admissible-interior uniqueness, transport, localization, and bookkeeping-collapse constraints.

Proof. Let an ensemble embedding be proposed. If all embedded members preserve the same standing-bearing classifications, witness ledgers, continuation behavior, quotient structure, and transport roles, then they are extensionally the same admissible interior, giving (i). If their differences change only representation, notation, or coordinate presentation, then (ii) holds.

Assume instead that at least one difference is standing-bearing. If the difference is introduced by changing the comparison envelope after the fact, it is scope transport, giving (iii). If standing from one member is used to authorize standing in another without independent target gating and target closure, then (iv) holds. If the ensemble difference is carried by an admissibility-relevant index, parameter, measure coordinate, or boundary selector, then (v) holds by the no-parameterization result. If a rule generates membership, selects representatives, repairs failed standing, or authorizes by probability, typicality, symmetry, observer count, or explanatory power, then (vi) holds by operator exhaustion.

If the ensemble instead claims standing through a cross-domain or hybrid-domain layer, then that layer must be independently gated and translation-disciplined. Failure gives the cross-domain obstruction. If the ensemble claims to compare distinct members by transport, then those transports must be explicitly licensed, mutually compatible, and non-divergent on overlaps. Failure gives the transport and compatibility obstruction. If the ensemble relies on replicated or locally indistinguishable members, then the multiplicity must be locally witnessed, finite or quotient-stabilized, and transport-closed; otherwise the footprint or replication obstruction applies. If the ensemble uses persistence through traces, branches, histories, or records, then continuation-locus structure is required. If it depends on observer, record, or realized-persistence predicates across members, then those predicates must be bounded-localized and transported under license. By Lemma 16.1, these cases exhaust the ways a formal family can claim inherited independent standing-bearing multiplicity. Thus multiplicity alone cannot confer standing. If all required structures are explicitly supplied, the construction is not inherited multiplicity; it is a newly gated target domain evaluated under the same closure constraints. $\square$

Corollary 16.4 (Admissible interior uniqueness corollary). *Within any shared admissibly embedded comparison regime, all members whose standing is inherited by that regime share one admissible interior up to extensional equivalence. A non-equivalent difference is bookkeeping, scope change, inadmissible, or part of a newly gated target domain evaluated internally rather than inherited from the original multiplicity.*

Corollary 16.5 (No inherited multiverse standing). *Multiplicity of domains, vacua, branches, or modal alternatives does not by itself confer admissible standing-bearing ensemble structure. Standing-bearing comparison must be independently licensed.*

16.1 The collapse is finite

The theorem is not an open-ended suspicion. It is an exhaustion result. Table 2 records the forced outcomes.

Table 2: Ensemble embedding collapse ledger.

Attempted route	Forced outcome
Same-domain plurality	Extensional collapse or bookkeeping
Alternative standing classifier	No competing classifiers; selector failure
Parameter-indexed alternatives	Boundary parameterization or inert index
Conservative extension	Definitional bookkeeping
Non-conservative extension	New domain or inadmissible repair/generator
Measure or typicality generation	Carrier laundering unless downstream of admitted ensemble
Cross-domain standing import	Standing transfer without target gating
Transport comparison	Requires explicit license and fixed domain
Composed transport comparison	Requires compatibility license
Cross-domain hybrid comparison	Requires independent target gate and regime translation discipline

Overlap disagreement	Requires illicit arbitration, privilege, or target change
Continuous parameter sweep	Holonomy obstruction unless quotient-stabilized
Trace-only continuation	Lacks continuation locus and does not carry standing-bearing persistence
Unbounded identical replication	Redundancy, transport incoherence, or finite envelope input
Globally inequivalent but locally indistinguishable family	Requires admissible footprint witness or loses standing-bearing multiplicity
Global observer class	Fails bounded localization or becomes explicit localized bridge
Diffuse persistence class	Fails stable consequence retention under perturbation
Landscape or branch multiplicity	Requires ensemble, transport, measure, and predicate licensing

17 Extensional Plurality versus Admissible Plurality

The theorem preserves several forms of plurality. This section prevents a common misreading.

Definition 17.1 (Extensional or presentation plurality). *Extensional or presentation plurality includes multiple descriptions, coordinate systems, gauges, notations, model presentations, definitional extensions, and representational skins that preserve the same standing-bearing content.*

Definition 17.2 (Admissibility-bearing plurality). *Admissibility-bearing plurality is plurality that changes standing-bearing classification, target fixation, witness compatibility, legal continuation, observer-predicate content, or admissible transport in a governance-relevant way.*

Theorem 17.3 (Plurality separation theorem). *Extensional or presentation plurality is compatible with the ensemble embedding collapse theorem. Admissibility-bearing plurality is not, unless it is relicensed as a new domain or collapses into one of the outcomes of Theorem 16.3.*

Proof. Presentation plurality changes no standing-bearing content and so survives as bookkeeping. Admissibility-bearing plurality changes at least one standing-relevant component. The main theorem exhausts the routes by which such change may be embedded. It either collapses extensionally, changes domain, or fails admissibility. Hence the two pluralities have different status. $\square$

Corollary 17.4 (Bookkeeping survival). *If a distinction changes no admissibility-bearing consequence, no witness compatibility, no lawful transport behavior, no quotient identity, and no bounded persistence structure, then it may survive as bookkeeping or presentation plurality without constituting an independent admissible interior.*

Proof. Such a distinction is neutral at all standing-bearing loci used by the theorem. It therefore triggers none of the collapse routes except the bookkeeping outcome. Bookkeeping survival is not a defect of the theorem; it is the intended residue of extensional plurality. $\square$

Remark 17.5 (No metaphysical monism). *At this point, the theorem is best read in its precise form. The result does not identify all descriptions. It identifies the standing-bearing admissible interior under the relevant closure conditions. Description-level plurality remains open inside that interior.*

18 Implications for Anthropic, Landscape, and Branch Ensembles

The paper’s immediate motivation lies in anthropic, landscape, multiverse, and branch-variation arguments. This section records the consequences without overextending them.

18.1 Anthropic ensembles

Anthropic reasoning requires a parameter space, a measure or weighting rule, an observer predicate, and transport of those objects across alternatives. Paper 5 established that anthropic conditioning is a posterior filter, not a primitive generator. The present result strengthens the background: the ensemble itself does not inherit admissibility-bearing structure by being named. It must pass embedding discipline.

Corollary 18.1 (Anthropic ensemble corollary). *An anthropic ensemble is admissible only to the extent that it supplies independent parameter-space typing, measure authority, observer-predicate localization, witness preservation, and transport compatibility. If those structures are absent, anthropic conditioning has no admissible standing-bearing target.*

18.2 Landscape ensembles

A landscape may define vacua, compactifications, effective theories, or low-energy parameter assignments. The present theorem does not prohibit such models. It says that multiplicity is not enough. Explicit landscape measures and vacuum ensembles are admissible if they are supplied by the model as part of an independently gated translation layer. In that case the explanatory force derives from that additional ensemble structure, not from multiplicity or anthropic conditioning alone. Landscape comparison must supply ensemble membership rules, measures, low-energy transport, witness bridges, observer predicates, overlap non-divergence, and compatibility conditions. If it does, the explanation is conditional on that explicit structure. If it does not, the ensemble does not carry standing-bearing explanatory authority.

18.3 Everettian branch structures

Everettian outcome branching under a fixed Hamiltonian produces outcome multiplicity, not automatic law or constant multiplicity. If branch variation is used to vary constants, the construction has moved from ordinary outcome branching to law-varying ensemble structure. That move requires the same target gating, transport, compatibility, and measure discipline as any other ensemble.

18.4 Modal parameter sweeps

Modal parameter sweeps often write a family of constant assignments and evaluate counterfactual consequences. Such sweeps can be useful inside a model. They become standing-bearing only when their parameter dimensions, transport, observer predicates, and measures are admitted. Otherwise they remain heuristic or presentation-level.

19 Worked Examples

The theorem is abstract, but its use is straightforward. The examples below are not empirical refutations of any model. They show how ensemble claims are role-classified.

19.1 Fine-structure-constant scan

A simple anthropic scan may write

$$\alpha \in (0, 1)$$

and ask which values permit atoms, chemistry, stars, or observers. The interval is not yet an admissible ensemble. A low-energy value of α is a dimensionless quantity, but dimensionlessness removes unit arbitrariness; it does not supply ensemble standing. The scan must specify which scale is meant, how running or matching is handled, which matter content is held fixed, which observer or chemistry predicate is transported, and which measure weights the interval.

The normal-form classification is as follows.

Component	Role under the theorem
Coordinate interval for α	Formal parameter presentation; not yet ensemble standing
Scale-indexed value $\alpha(\mu)$	Transport-defined or effective quantity; requires scale and scheme context
Low-energy measured value	Boundary/envelope input inside an admitted physical domain
Life-permitting subset	Observer/persistence predicate; requires transport and localization
Uniform prior over interval	Measure choice; not generated by the interval itself
Chemistry/atom predicate	Witness predicate; must remain comparable across the scan

Thus the scan may be a useful model. It is not a primitive standing-bearing ensemble unless the missing structure is supplied.

19.2 Cosmological constant anthropic bound

Anthropic reasoning about the cosmological constant is often more disciplined than generic parameter scanning because the observer proxy may be tied to structure formation. The present theorem does not deny that such reasoning can be conditionally informative. It classifies the dependencies.

A conditional bound on Λ requires at least: a prior or weighting over Λ , an ensemble of domains or histories over which Λ varies, a structure-formation criterion, a proxy connecting structure to observer or record-bearing standing, and a bridge from the ensemble to our observation. If those structures are explicit, anthropic reasoning survives as posterior filtering. If the anthropic condition is asked to supply them, the construction instantiates the collapse theorem.

A well-specified anthropic bound can remain conditionally informative. The important point is not that the bound is useless. It is that its explanatory authority is conditional on the ensemble structure. The observer condition does not generate the measure, the measure does not generate standing, and the ensemble does not inherit comparability merely by containing alternatives.

19.3 String-landscape-style EFT vacua

A landscape of vacua may contain many low-energy effective descriptions. Such a landscape is not automatically a same-domain parameter scan. Vacua may differ in compactification data, spectrum, gauge structure, symmetry breaking, moduli stabilization, and low-energy EFT content. Some differences may be presentation-level; others may be scope-level or anchor-level.

For landscape reasoning to be standing-bearing, the model must specify: what counts as a member, how vacuum identity is fixed, how low-energy constants are transported, how measures are assigned, how observer proxies are compared, and how empirical witness structure connects the ensemble to observation. If all of this is supplied, the landscape is a retyped target domain with its own gate. If not, multiplicity remains formal or collapses into one of the inadmissible routes.

19.4 Everettian outcome branching

Everettian branching under a fixed Hamiltonian inherits the same law and constant structure. Outcome multiplicity is therefore not constant multiplicity. If a branch family is used to support anthropic conditioning over outcomes under fixed law, the relevant task is branch measure and record localization. If it is used to scan laws or constants, a new law-varying ensemble has been introduced. That ensemble requires fresh target gating and transport compatibility. The branch structure does not automatically license it.

19.5 Generic modal worlds

A possible-world construction may be useful for counterfactual reasoning. But a physical explanatory ensemble requires more than modal description. It requires typed alternatives, standing-preserving comparison, witness predicates, and a measure if probability is claimed. If a modal world differs only representationally, it is not a distinct standing-bearing interior. If it differs governance-relevantly, it requires target gating or fails admissible embedding.

20 No Metaphysical Overclaim

Because the theorem is strong, its non-claims must be explicit. The theorem is not a denial of formal plurality, representational plurality, model plurality, or modal discourse. It concerns admissibility-bearing governance structure under fixed-domain continuation and cross-domain translation discipline.

Proposition 20.1 (No world-uniqueness entailment). *The Ensemble Embedding Collapse Theorem does not entail that only one metaphysical world exists.*

Proof. The theorem quantifies over admissible standing-bearing ensemble embeddings. It does not quantify over metaphysical worlds as primitive entities. It classifies what happens when such purported worlds are used as standing-bearing members of an admissible comparison regime. Therefore no world-uniqueness thesis follows. $\square$

Proposition 20.2 (No denial of formal plurality). *The theorem does not deny formal plurality, model plurality, or exploratory counterfactual construction.*

Proof. Formal plurality can exist without standing-bearing ensemble authority. The theorem constrains only the promotion of formal alternatives into admissible comparison members. It leaves formal study and representation-level plurality untouched. $\square$

Proposition 20.3 (No empirical prediction). *The theorem introduces no new empirical dynamics, equations, cosmological models, or physical predictions.*

Proof. The result is structural. It classifies the admissibility conditions required for ensemble comparison. It does not modify physical laws or add dynamical content. $\square$

21 Objection Matrix

Objection	Response
This is modal collapse.	No. Modal and formal alternatives may still be described. The theorem concerns standing-bearing admissible comparison structure.
This is metaphysical monism.	No ontology of one substance, one world, or one material reality is asserted. The theorem is about admissible interiors and embedding.
This bans multiverse models.	No. It requires multiverse models to supply transport, compatibility, measure, observer-predicate, and witness structure rather than inheriting them by multiplicity.
This is definitional.	No. The collapse follows from finite routes: no boundary parameterization, no competing classifiers, no standing-extending operators, no automatic transport, no automatic compatibility, and no diffuse persistence.
This is anti-physics.	No. The theorem does not modify physics. It constrains explanatory use of ensemble comparison.
Landscapes can define measures.	Yes. If a model supplies a measure, its claims are conditional on that structure. The measure is not generated by anthropic or ensemble multiplicity alone.
Observers can be defined functionally.	Functional definitions are admissible only if their standing-bearing predicates are transported, localized, and witness-compatible across the ensemble.
Everett gives many branches.	Outcome branches under fixed law do not automatically produce standing-bearing law or constant variation. Law-varying branch ensembles require additional structure.
Parameter scans are useful.	Yes. Usefulness is not the target. The theorem addresses when a scan becomes an admissible standing-bearing explanatory ensemble.
The result is too restrictive.	It is restrictive only for standing-bearing authority. Bookkeeping, formal construction, representation, and exploratory modeling remain available.
Cross-domain hybrids evade same-domain closure.	Only if the hybrid is independently typed, gated, transported, and closed as a target domain. Otherwise it is scope transport or cross-domain laundering.
Infinite ensembles are common in physics.	Formal infinite families may be useful. The theorem targets infinite standing-bearing multiplicity, especially unbounded standing-indistinguishable duplication or locally indistinguishable footprint multiplicity.
The theorem assumes uniqueness.	No. Uniqueness is the residue after finite route exhaustion: no parameterization, no competing classifiers, no transfer, no totalizing extension, no unlicensed transport, no trace-only continuation, no unbounded duplicate multiplicity, and no diffuse persistence.

Continuation is just trace succession.

Trace succession may be a valid formal record, but standing-bearing continuation requires a preserved continuation locus.

22 Conclusion

The theorem proved here is an ensemble theorem. It does not say that all descriptions are one, that formal plurality disappears, or that only one metaphysical universe exists. It says that governance-relevant admissibility-bearing plurality fails to survive the joint constraints of fixed-domain standing, no admissibility-relevant parameterization, classifier uniqueness, operator exhaustion, cross-domain closure, regime translation discipline, transport licensing, compatibility licensing, continuation-locus preservation, multiplicity minimality, and bounded persistence localization.

What remains is not the absence of plurality, but standing-preserving admissible structure whose governance-relevant content is unique up to extensional equivalence. Plural descriptions may persist as presentations, coordinate systems, gauges, models, heuristics, or definitional extensions. But when plurality is asked to carry standing-bearing explanatory force, it must pass the embedding test. The test is finite: it either preserves the same interior, remains bookkeeping, changes scope, imports standing, relies on selector or carrier structure, fails cross-domain translation, fails transport or compatibility, lacks continuation-locus preservation, relies on inadmissible multiplicity, or lacks bounded localization.

The result is therefore not an argument against plurality as such. It is a closure theorem for inherited admissibility-bearing ensemble standing. In that sense, ensemble collapse is the natural culmination of the constants arc: fixed-domain constants cannot be freely varied, constant-like quantities must be role-classified, running is lawful transport rather than arbitrary variation, parameter mobility is role-dependent, anthropic conditioning is posterior rather than primitive, and ensemble multiplicity itself does not inherit standing by being named. What survives is not the absence of plurality, but standing-preserving admissible structure whose governance-relevant content is unique up to extensional equivalence within admissibility-bearing governance.

A Theorem Dependency Ledger

Result used in this paper	Source stream
No admissibility-relevant parameterization	AMetric boundary; Unus Solus Possibilis Est; Standing-Admissibility Identity Closure
No competing classifiers	Standing-Admissibility Identity Closure; admissible interior uniqueness
No repairs, generators, or carriers	Closure by Exhaustion; Unus Solus Possibilis Est
No standing transfer	Standing-Admissibility Identity Closure; scope-discipline arc
No admissibility-preserving extension	Closure by Exhaustion via partial algebras
Transport licensing	GR-7 minimal licensed transport schema
Transport compatibility licensing	GR-9 stability and compatibility of transport
Continuous parameter transport obstruction	Transport-Closure Locus for Continuous Parameters

Bounded persistence localization	Standing Localization under Realized Persistence
Continuation-locus induction	Continuation-Locus Induction under Admissibility-Bearing Governance
Conditional footprint multiplicity strengthening	Nondegenerate Standing Forbids Infinite Footprint Multiplicity (SND+), used only conditionally and not as a main-proof dependency
Generation multiplicity minimality	Generation Multiplicity Minimality and Non-Degeneracy
Cross-domain closure	Unified Closure Certificate; Cross-Domain Closure Certificate Unification
Regime translation discipline	Regime Translation Layer: Cross-Domain Lift
Anthropic posterior survival	The Anthropic Collapse Theorem
Parameter mobility and running discipline	Papers 1-4 of the constants arc

B Claim Index

Claim	Where established
A formal family is not yet a standing-bearing ensemble.	Definitions 4.6 and 4.7; Figure 1
Boundary indexing does not create admissible alternatives.	Theorem 5.2; Corollary 5.3
Same-domain classifier disagreement collapses.	Theorem 6.2
Governance-role distinction without classification or trajectory effect is bookkeeping.	Theorem 6.3
Same-domain ensemble plurality collapses or becomes presentation-level.	Corollary 6.4
Formal, inherited, and explicitly gated multiplicities are distinct.	Definitions 2.2–2.4
Extensions are definitional, scope-changing, or inadmissible.	Theorem 7.3
Primitive ensemble generation is exhausted by four obstruction families.	Theorem 10.2
Ensemble multiplicity routes are exhaustive.	Lemma 16.1
Transport is not automatic.	Theorem 11.2
Transport compatibility is not automatic.	Theorem 11.4
Continuous parameter families face transport-holonomy obstruction.	Theorem 11.5
Standing-bearing persistence localizes under repair semantics.	Theorem 13.3

Observer-predicate embedding is not global by default.	Theorem 14.3
Cross-domain escape requires independent target gating.	Theorem 8.2
Overlap divergence requires illicit arbitration or target change.	Overlap non-divergence lemma in Section 8
Continuation-bearing ensemble use requires a continuation locus.	Continuation-locus induction in Section 12
Unbounded indistinguishable replication cannot carry ensemble depth.	No unbounded standing-indistinguishable replication theorem in Section 12
Admissible ensemble embedding has a finite normal form.	Definition 15.1; Proposition 15.2
Inherited independent standing-bearing ensemble multiplicity is not conferred by multiplicity alone.	Theorem 16.3
Extensional plurality is preserved.	Theorem 17.3; Corollary 17.4

C Short Proof Form

1. A standing-bearing ensemble must support admissible comparison among its members.
2. Admissible comparison requires standing-preserving transport, compatible composed transports, and witness preservation.
3. Boundary-positive alternatives cannot be admissibility-relevantly indexed without introducing forbidden parameterization or selectors.
4. Same-domain standing classifiers cannot disagree; any governance-relevant disagreement requires a forbidden selector or produces indeterminate standing.
5. Admissibility-preserving extensions are definitional, scope-changing, or inadmissible.
6. Primitive generators, measures, typicality rules, canonical representatives, and repairs reduce to boundary selectors, generators, carriers, or scope transport.
7. Cross-domain or hybrid-domain comparison requires an independently gated target layer and overlap non-divergence.
8. Continuation-bearing ensemble use requires a continuation locus; trace-only succession is insufficient.
9. Unbounded standing-indistinguishable replication does not supply standing-bearing ensemble depth; SND+ provides only conditional support for the stronger footprint-multiplicity claim.
10. Observer or record predicates used standing-bearingly across an ensemble require bounded localization or explicit transport.

11. Therefore every proposed inherited independently standing-bearing ensemble either collapses extensionally, remains bookkeeping, changes domain, imports standing, fails transport or compatibility, uses forbidden selector/carrier structure, fails continuation-locus preservation, relies on inadmissible multiplicity, or fails bounded localization.
12. Hence no inherited independently standing-bearing ensemble multiplicity is conferred by multiplicity alone; surviving multiplicity is explicitly gated, internally governed, or bookkeeping.

References

- [1] Amos Jay Maley, *The Standing Fixity of Physical Constants*, 2026.
- [2] Amos Jay Maley, *A Classification Theorem for Physical Constants*, 2026.
- [3] Amos Jay Maley, *Renormalization as Standing-Preserving Scale Transport*, 2026.
- [4] Amos Jay Maley, *Why Some Parameters Run and Others Cannot*, 2026.
- [5] Amos Jay Maley, *The Anthropic Collapse Theorem*, 2026.
- [6] Amos Jay Maley, *Standing-Admissibility Identity Closure*, 2026.
- [7] Amos Jay Maley, *Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior*, 2026.
- [8] Amos Jay Maley, *Closure by Exhaustion of Standing-Bearing Operator Regimes*, 2026.
- [9] Amos Jay Maley, *GR-7: Minimal Licensed Transport Schema*, 2026.
- [10] Amos Jay Maley, *GR-9: Licensed Stability and Compatibility of Transport*, 2026.
- [11] Amos Jay Maley, *Transport-Closure Locus for Continuous Parameters*, 2026.
- [12] Amos Jay Maley, *Standing Localization under Realized Persistence*, 2026.
- [13] Amos Jay Maley, *ARC-5: Commitment Scope and Illicit Scope Transport*, 2026.
- [14] Amos Jay Maley, *The AMetric Boundary*, 2026.
- [15] Amos Jay Maley, *Continuation-Locus Induction under Admissibility-Bearing Governance*, 2026.
- [16] Amos Jay Maley, *Nondegenerate Standing Forbids Infinite Footprint Multiplicity (SND+)*, 2026.
- [17] Amos Jay Maley, *Generation Multiplicity Minimality and Non-Degeneracy*, 2026.
- [18] Amos Jay Maley, *Unified Closure Certificate: Cross-Domain Admissibility – Certified Closed by Exhaustion*, 2026.
- [19] Amos Jay Maley, *Cross-Domain Closure Certificate Unification*, 2026.
- [20] Amos Jay Maley, *Regime Translation Layer: Cross-Domain Lift*, 2026.